Determination of pump NPSH margin for safe operation

Index

1. Purpose 3
2. Scope 3
3. Definitions 3
4. General Procedure - Description 4
5. Description of the methodology 5
   5.1 ANSI/HI [1] std. 5
   5.2 Gulich methodology [2] 7
   5.3 TMP handbook [3] 9
   5.4 Internal procedure for severe applications 10
6. Description of the module Mod.533 14
7. Acceptance criteria 17
8. Limitations 18
9. References 18

1. Purpose

This document establishes the criteria to calculate the safety margin calculation of Net Positive Suction Head (NPSH) required by using *ANSI/HI [1]*, *Gulich [2]* and *TMP [3]* formulations.

2. Scope

This procedure is applicable to both single and double suction first stage impeller, single- or multi-stage pumps.
The procedure is applicable to the plant of Nova Milanese and the plant of La Spezia.

3. Definitions

n	rotational speed of the pump rotor	rpm
Qrtd	flowrate at rated condition	m^3/h or GPM
Hrtd	pump head at rated condition	m or ft
Hst	head per stage	m or ft
Qbep	flowrate at best efficiency point	m^3/h or GPM
Pstage	power per stage at best efficiency point	kW or HP
NPSHav	available net positive suction head	m or ft
AOR	allowable operating region (as per [1])	
POR	preferred operating region (as per [1])	
NPSHr	required net positive suction head	
NPSHr3%	required net positive suction head for 3% head drop	m or ft
NPSHR	recommended net positive suction head	m or ft
R	ratio between NPSHR and NPSHr	

DELTA margin between NPSHR and NPSHr

Specific speed Nq or Ns

Suction specific speed sss or Nss

U1 peripheral velocity at impeller D1 m/s or ft/s

4. General Procedure - Description

The evaluation of pump NPSHr safety margin is calculated using the **Mod.533** spreadsheet.

This method considers of the following pump features:

- Single/double suction 1st stage impeller
- Mixed flow pump
- n
- Qrtd
- Hrtd
- Hst
- Qbep
- Pstage
- NPSHav
- D1tip
- Pump service (boiler feed, condensate, cooling tower, circulating, water, wastewater)
- Plant application (hydrocarbons, chemical, electric non-nuclear, water/wastewater API)
- Impeller material (cast iron, stainless, aluminium, bronze)
- Water ph
- Ppm of H2S
- Fluid temperature
- Special fluids (i.e. kerosene, petrol, light fuel oil, methanol, butane, i-butane)
- NPSHr3% curve

5. Description of the methodology

The calculation of the NPSH safety margin defines the value of NPSHR to apply to guarantee a proper operation life of the pump.

The selected pump must accomplish the following relations:

- NPSHav>NPSHR
- NPSHR= NPSHr*R

  or
- NPSHR=NPSHr+DELTA

The method is based on different formulations described below:

5.1 ANSI/HI [1] std.

The approach evaluates the safety ratio R(= NPSHR/NPSHr) and safety margin DELTA(=NPSHR-NPSHr) expressed in m or ft, for different type of pump applications. Two different values are calculated for the AOR and POR

<u>For hydrocarbon process</u>:

R=1.1 or

DELTA=1 m (3.3ft)

Within AOR and POR

<u>For chemical process</u>:

for sss<210 (11000 NSS)

R=1.1 or

DELTA = 0.6m (2ft)

within AOR

for sss>210 (11000 NSS)

R=1.1 or

DELTA = 0.6m (2ft)

within POR

R=1.2 or

DELTA = 1 m (3.3ft)

within AOR, POR excluded.

For electric power plant process (non-nuclear) see the table below from *Chapt. 9 References [1].*

Table 9.6.1.4.3 — NPSH margin, electric power plant pumps (non-nuclear)

Pump Type	Power (kW/stage)	POR NPSH Ratio	AOR NPSH Ratio/Margin
Boiler feed	< 225	1.1 ratio	1.3 ratio
Boiler feed	≥ 225 and < 500	1.2 ratio	1.5 ratio
Condensate	All	1.0 ratio	1.0 ratio
Cooling tower	All	1.1 ratio	1.3 ratio
Circulating/Cooling water	All	1.05 ratio	1.0 m

for water and wastewater process see the table below *Chapt. 9 References [1].*

Table 9.6.1.4.5 — NPSH margin, water/wastewater pumps

Pump Type	Power (kW/stage)	POR NPSH Margin (1.0 m minimum)	AOR NPSH Margin (1.5 m minimum)
Wastewater (cast-iron impeller)	< 45	1.1 ratio	1.2 ratio
Wastewater (stainless-steel impeller)	< 45	1.05 ratio	1.1 ratio
Wastewater (cast-iron impeller)	≥ 45	1.2 ratio	1.3 ratio
Wastewater (stainless-steel impeller)	≥ 45	1.1 ratio	1.2 ratio
Water (stainless or aluminum–bronze impeller)	< 75	1.05 ratio	1.1 ratio
Water (stainless or aluminum–bronze impeller)	≥ 75	1.1 ratio	1.2 ratio

<u>For slurry process:</u>

R=1.1 or

DELTA=0.6 m (2ft)

Within AOR and POR

5.2 Gulich methodology [2]

The methodology is based on the formulation proposed by *Chapt. 9 References [2]* in *Table 6.2.* The method evaluates the safety factor on NPSHr3% for all centrifugal pumps. The description of the formulation is shown on *Table 6.2:*

Table 6.2 Selection of available $NPSH_A$ to ensure safe operation

Purpose: this table allows to determine the available $NPSH_A$ necessary to ensure trouble-free operation. The following criteria are used:

- C1: Ensure that pump can run in the entire range of operation, in particular at maximum flow
- C2: Limit cavitation induced noise and vibration
- C3: Avoid (excessive) cavitation erosion

u_1 (m/s)	Fluid	Criteria	Determination of the necessary $NPSH_A$	
< 10	All	C1 (C2)	$NPSH_A \geq 1.25 \times NPSH_3$ Minimum: $NPSH_A = NPSH_3 + 0.6$ m	
< 20	Hydrocarbons (except kerosene and similar)		Objective: ensure operation range; cove tolerances and uncertainties	
10 to 50	Water	C2, C3	$\sigma_A \geq \sigma_3 + F_R F_F \{0.05 + 2 (u_1/u_{Ref})^3\}$ $u_{Ref} = 100$ m/s. Apply only for $\sigma_A < \sigma_i$	
			Fluid factors	**F_F**
			Water with $\alpha > 5$ ppm; $t < 200$ °C	1.0
			Sea water or other corrosive media	≥ 1.15
			Hydrocarbons with $\Delta T_u > 5$ °C	0.75
20 to 60	Hydrocarbons		**Risk factors**	**F_R**
			Operation at $0.8 < q^* < 1.1$	1.0
			Operation at partload recirculation or pressure surface cavitation at $Q > Q_{SF}$	1.2
			Non-uniform approach flow	≥ 1.1
50 to 75	Water Cr-Ni-steel	C3 C1	Cavity length, cavitation noise or operation experience under the *same condition,* must be available for checking the impeller life according to Table 6.1	
> 60	Hydrocarbons Cr-Ni-steel	C1 C2	Cavity volume must be limited in order to avoid excessive vibrations, noise and pulsations	
> 75	Water at T < 200 °C Cr-Ni-steel	C1 C3	$NPSH_A \geq NPSH_i$ Practically bubble-free operation is necessary; otherwise damage after short periods of operation.	
10 to 25	De-aerated water $\alpha < 5$ ppm, T < 100 °C Cr-Ni-steel	C2	$\sigma_A \geq \sigma_3 + F_R F_F \{0.05 + 2(u_1/u_{Ref})^3\}$ $u_{Ref} = 100$ m/s	
> 25		C3	Very erosive; cavity volume or cavitation noise must be limited, Table 6.1	

Minimum: $NPSH_A \geq 1.25 \times NPSH_3$ or $NPSH_A = NPSH_3 + 0.6$ m whatever is greater

The safety factor on $NPSH_3$ is defined as: $F_{NPSH} = \frac{NPSH_A}{NPSH_3} = \frac{\sigma_A}{\sigma_3}$

The available NPSH determined in this way is only sufficient if the following is observed:

1. The impeller is designed for an incidence at BEP i < 3° (to 5°).
2. The approach flow is reasonably uniform (criteria are given in Chap. 11.7.3)
3. There is no excessive vortex formation.
4. The selected $NPSH_A$ is also sufficient for the maximum continuous flow.

Caution: The above procedure is empirical and can give no guarantee in extreme cases. It may be conservative in other instances.

Table 6.2

5.3 TMP handbook [3]

The approach provides a NPSH safety margin calculation based the inlet diameter rotating speed, as indicated in the *Table C.8.1.*

	definition	limitation	area of application
(a)	$NPSH_{av} \geq NPSH_{3\%} + 0{,}5$ m	u1 < 10 m/s	small size pumps
(b)	$NPSH_{av} \geq SA \cdot NPSH_{3\%}$ SA = 1,25	u1 < 20 m/s	medium size pumps
(c)	$NPSH_{av} \geq SA \cdot NPSH_{3\%}$ SA (see Fig.C.8.1.)	u1 < 60 m/s	general criteria for industrial pump application
(d)	$NPSH_{av}$ determined on the basis of equations from Table C.8.2.	u1 > 60 m/s H (stage) > 600 m	pumps of high power concentration

Table C.8.1 Criteria and cavitation safety values

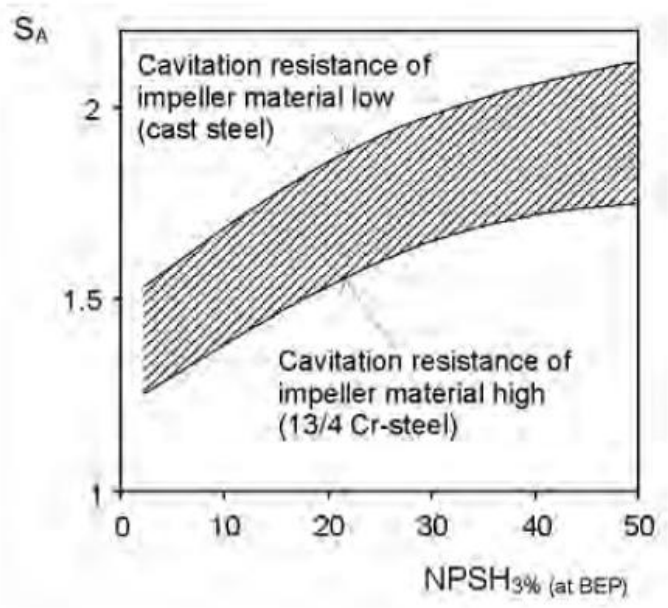


Fig.C.8.1 Cavitation safety values SA as a function of $NPSH_{3\%}$ (at BEP)

Table C.8.1

The procedure is applicable for pumps belonging to categories (a), (b) and (c) of *Table C.8.1.*

Pumps falling on category (d) of *Table C.8.1.*, requires a special treatment based on erosion rate that is out of the scope of this document.

5.4 Internal procedure for severe applications

The method is applicable to pumps handling high corrosive fluids (fluids with acid ph or fluids with consistent ppm of Sulphuric acid).

The formulation used is described below:

*NPSHr =Sn*Sen*St*Sm*Slg*NPSH3% and NPSHr > NPSHr3% + 0.5m (1.6ft)*

Where:

- **Sn = NPSHr0%/NPSHr3%** is calculated by the diagram below:

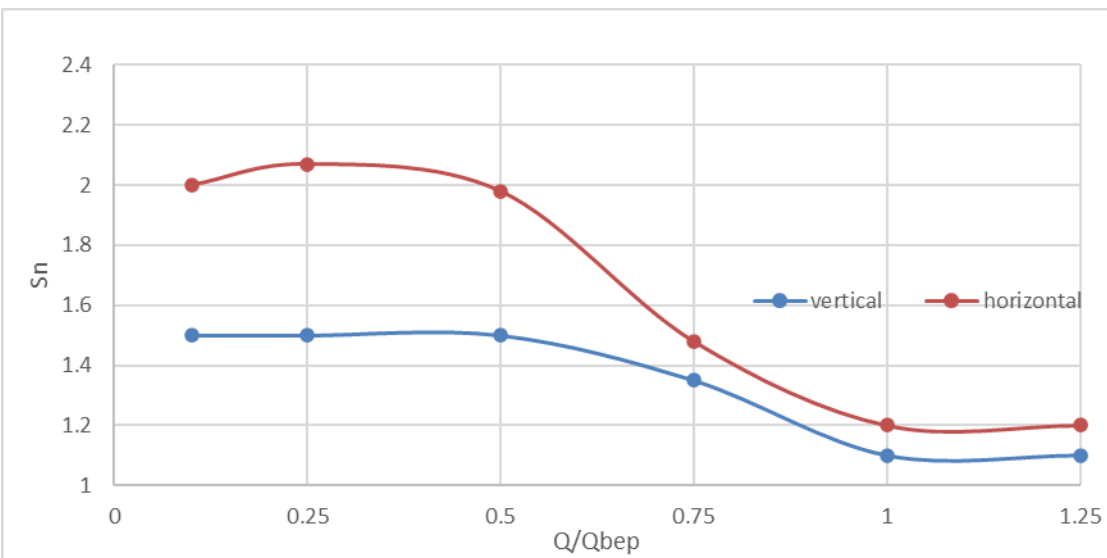


Figure 1. diagram for Sn

- **Sen** is a factor based on cavitation energy:

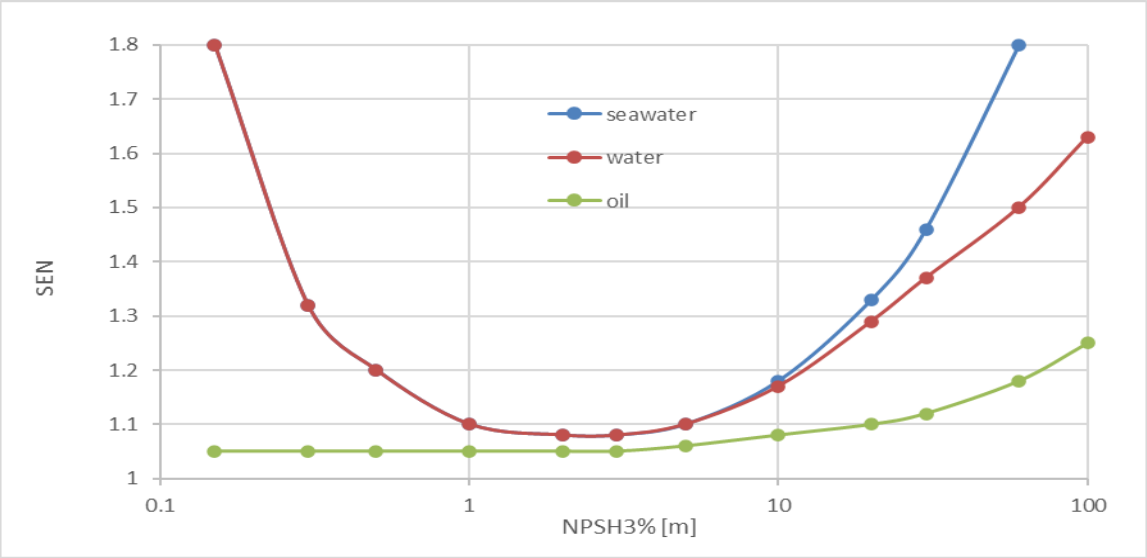


Figure 2 Diagram for Sen

- **Sm** is based on the material of the impeller.

  Sm values are listed in the two tables below:

Table 1 Sm for water and seawater

material	water ph<6	water ph>6	seawater or H2S<1ppm
stainlees	1	1.05	1.1
stainlees	1	1.05	1.1
stainlees	1	1.05	1.1
stainlees	1	1	1
stainless alu bronze	1	1	1
stainless alu bronze	1.05	1.05	1
stainless alu bronze	1.05	1.05	1.05
cast iron	1.05	1.15	
cast iron	1.05	1.15	
cast iron	1.1	1.2	

Table 2. Sm for special fluid			
chem comp	H2S	ph4	ph6
G-X20 Cr 14	1	1.1	1.05
G-X5 Cr Ni Mo 134	5	1.12	1.05
G-X22 Cr Ni 17	10	1.14	1.06
G-X6 Cr Ni Mo 1810	20	1.18	1.07
G-NI Al Bz F 60	30	1.22	1.08
G-Sn Bz 10	40	1.265	1.09
G-Cu Al 10 Ni	50	1.3	1.1
GGG 40	60	1.3	1.1
SS-C25	500	1.3	1.1

- **Slg** is based on the customer request to fulfil the impeller life guarantee of 40000hr . in this case Slg=1.2, otherwise Slg=1.
- **St** is based on the effect of temperature on NPSHr3% measured in fresh water:

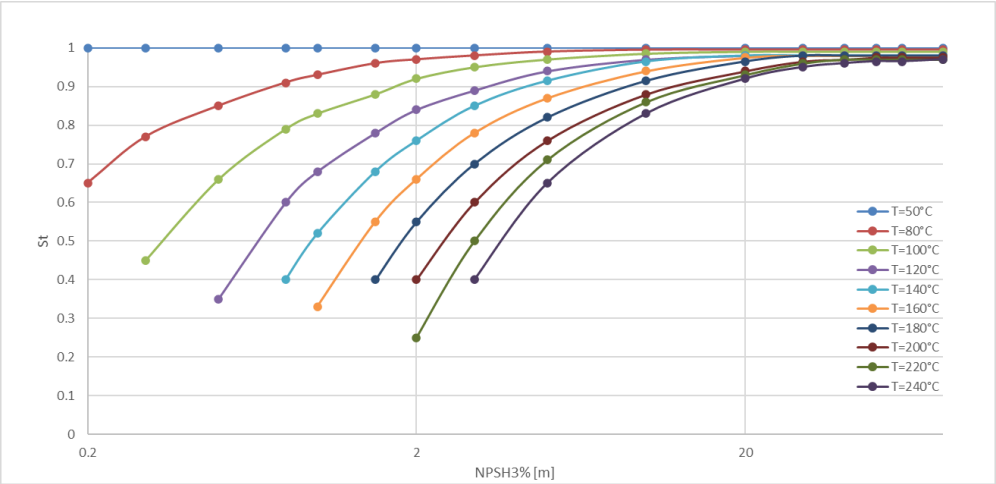


Figure 3. Diagram for St

For hot fluid other than water, the value of temperature to be considered in the diagram above is calculated adding a delta value based on *Table 3*.

Table 3. delta temperature for other fluids

fluid	dT [°C]
kerosene	0
petrol	0
light fuel oil	0
methanol	50
butane	120
i-butane	140
propane	180
freon	90

6. Description of the module Mod.533

The mask for the input data is shown in *Figure 4.*

Only data highlighted in the grey cells must be compiled.

Tested required NPSHr3% must be added in the **Mod.533.**

PUMP MODEL	
PUMP SIZE	
DRAWING NUMBER	

INPUT DATA		
unit system	SI	
Pump speed	324	RPM
Qrtd	53000	m3/h
Hrtd	27	m
Hst	27.0	m
Qbep	53000.0	m3/h
Pstage	3500.0	kW
NPSHav	14.7	m
D1tip	1000.0	mm
pump service	API	
plant application	hydrocarbon	
impeller material	ASTM A216 Gr WCB	

mixed flow pump	☐		
ph water		use for special application	
ppm H2S		use for special application	
fluid temperature	20	°C	
special fluid			

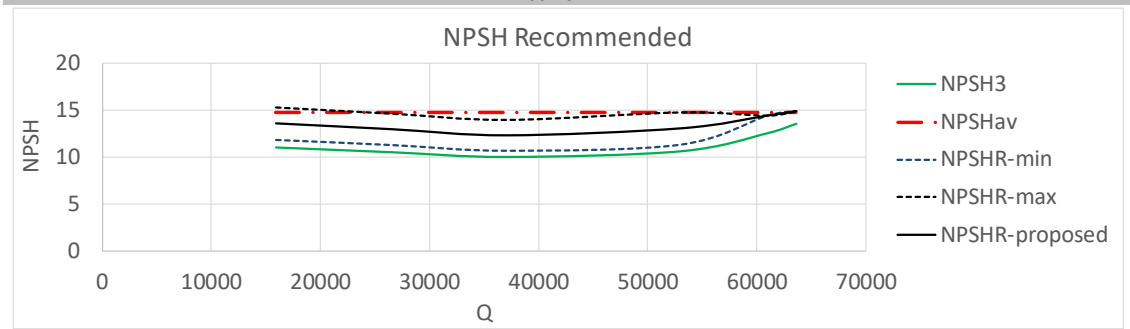
		Q/Qbep	NPSH3 m	Q m3/h
AOR		0.3	11	15900
		0.5	10.5	26500
	POR	0.7	10.0	37100
	bep	1	10.6	53000
		1.15	12.5	60950
		1.2	13.5	63600

Figure 4: Input data for NPSH Safety margin calculation

The outcomes of the spreadsheet are the NPSH safety margin (in terms of R and delta) calculated with the above-described methodologies. The last methodology described in *Chapt. 5.4* will activate only in case of pump handling corrosive fluids (ph<4 or ph>6 or H2S>1ppm) The program calculates the NPSH recommended using all the approaches and provides the minimum and maximum value in the range of desired flowrate.

The output data of the module are showed in *Figure 5.*

OUTPUT DATA



OUTPUT DATA

	R	
	AOR	POR
H.I. 9.6.1	1.1	1.1
J.F.Gulich	1.07	1.07
TMP-1	1.3898	1.39
TMP-2	2.0881	1.58

u1	16.96	m/s

	delta	
	AOR	POR
H.I. 9.6.1	1.0	1.0
J.F.Gulich	0.60	0.60
TMP-1	0.00	0.00
TMP-2	0.50	0.50

Q	HI	Gulich	TPI_standard	TPI_special	NPSHRmin	NPSHRmax	NPSHRave	Rave
15900.0	12.1	11.8	15.3		11.8	15.3	13.6	1.2
26500.0	11.6	11.3	14.6		11.3	14.6	12.9	1.2
37100.0	11.0	10.7	13.9		10.7	13.9	12.3	1.2
53000.0	11.7	11.3	14.7		11.3	14.7	13.0	1.2
60950.0	14.4	14.4	14.4		14.4	14.4	14.4	1.15
63600.0	14.9	14.9	14.9		14.9	14.9	14.9	1.10

Figure 5. Output data for NPSH Safety margin calculation

7. Acceptance criteria

The criteria used on the Proposed NPSHR calculation is summarized in the following Table:

NPSHR	Q≤ Qbep	Qbep<Q<Qmax
(1) Minimum	Maximum value between • NPSHr+0.5m (15.6ft) • NPSHr*1.1	Maximum value between • NPSHr+0.5m (15.6ft) • NPSHr*1.1
(2a) Maximum – standard applications	Maximum value between • NPSHr + maximum delta (from [1], [2] and [3] calculation) • NPSHr * R (from [1], [2] and [3] calculation)	Maximum value between • NPSHr+0.5m (15.6ft) • NPSHr*1.1
(2b) Maximum – severe applications	Maximum value between • NPSHr + maximum delta (from [1], [2] [3] and internal procedure for severe application calculation) • NPSHr * R (from [1], [2] and [3] calculation)	Maximum value between • NPSHr+0.5m (15.6ft) • NPSHr*1.1
Proposed	**Average value between** • **(1)** • **(2a) or (2b)**	**Maximum value between** • **NPSHr+0.5m (15.6ft)** • **NPSHr*1.1**

8. Limitations

The method is applicable for pumps with inlet tip speed of the impeller lower than 50m/s (164ft/s). Higher speed requires special calculation based on the evaluation of the erosion rate of the impeller.

The flow approaching on the impeller must be uniform, without any excessive vortex formation.

NPSH available must be sufficient to cover the maximum continuous flow.

The NPSH safety margin calculated by this tool fulfils the request of continuous operation, absence of cavitation noise and vibration, guarantee of component life. It gives too conservative result, it shouldn't be used, in case of short time operation or if the customer requires different values.

Low specific speed pumps, operating at low inlet tip speeds can operate normally at NPSH close to NPSH3%.

The application of the method for these pumps will give a conservative outcome.

Pump with high specific speed, with high inlet tip speeds are affected by noise and higher vibrations. In this case it's strongly recommended to use the most conservative margin.

An excessive NPSH margin can result in a non-optimal pump selection (big size slow speed pumps).

9. References

[1] ANSI/HI – 9.6.1-2012 Rotodynamic Pumps Guideline for NPSH Margin

[2] J.F.Gulich "Centrifugal Pumps" Fourth Edition - 2020

[3] TM.P. "Centrifugal Pump Handbook" - 2002